

SEMESTER PROJECT

Title: Next-Best-View Aerial Swarms for Multi-Viewpoint Monitoring**Student(s):** Advaith Sriram (Robotics)**Professor:** Dario Floreano**Assistant 1:** Benjamin Jarvis**Project description:****Problem**

Next-Best-View (NBV) is a planning approach that optimizes a sequence of viewpoints to efficiently observe an object or scene. This is widely used in single camera applications, and while there is some research into multi-agent NBV [1][2], it has not been coupled with swarming algorithms. Swarms allow multiple simultaneous perspectives of a scene, however, it is not understood how viewpoints can be chosen while maintaining cohesion.

Goal

Swarming algorithms coordinate multiple agents to distribute themselves in space according to a series of rules, such as avoiding collisions with other agents and obstacles. This spatial distribution bears similarities to NBV approaches. The goal of this project is to first assess how well swarming algorithms passively maximise scene visibility compared to established NBV approaches. We then want to investigate new methods to improve the scene visibility while maintaining a cohesive swarm.

Method

1. Literature review on suitable models and similar interfaces
2. Implement an existing NBV approach
3. Compare its performance to a swarm algorithm
4. Develop a new approach that augments the swarm algorithm

[1] Patel, IROS, 2023

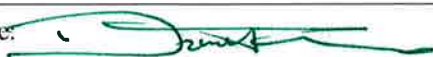
[2] Dhami, IROS, 2024

Remarks:

You should present a research plan (Gantt chart) to your first assistant before the end of the second week of the project. An intermediate presentation of your project, containing 8 minutes of presentation and 7 minutes of discussion, will be held on October 27, 2025 (all day). The goal of this presentation is to briefly summarize the work done so far and discuss a precise plan for the remaining time of the project. Your final report should start by the original project description (this page) followed by a one page summary of your work. This summary (single sided A4), should contain the date, laboratory name, project title and type (semester project or master project) followed by the description of the project and 1 or 2 representative figures. In the report, importance will be given to the description of the experiments and to the obtained results. A preliminary version of your report should be given to your first assistant at the latest 10 days before the final hand-in deadline. A PDF (dated & signed) of your final version should be sent to your first assistant and to the administrative assistant of the lab before noon January 8th, 2026 before noon. Failing this deadline will reflect in the final grade. A 20 minute project defense, including 5 minutes for discussion, will take place January 13th, 2026 (all day). You will be graded based on your results, report, final defense, and working style. All documents, including the report (source and pdf), summary page and presentations along with the source of your programs should be handed-in as a single compressed file on the day of the final defense at the latest.

Responsible professor:**Responsible assistant:**

Signature:



Dario Floreano

Signature:



Benjamin Jarvis

Next-Best-View Aerial Swarms for Multi-Viewpoint Monitoring

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SUMMARY

Objective: Multi-drone systems are increasingly used for autonomous scene monitoring and 3D reconstruction, where efficient viewpoint selection and coverage remain key challenges. This project presents a comparative study of decentralized swarm-based exploration and Next-Best-View (NBV) planning for multi-drone systems in simulated environments.

Methodology: The experiments were conducted in a Unity-based simulation environment, where multiple aerial drones equipped with depth cameras observed textured 3D house models from the House3K dataset. In the NBV planning, an information-driven strategy leveraging the PoinTr model for point cloud completion was used, iteratively selecting viewpoints that maximized information gain. Swarm-based exploration was implemented using periodic depth image acquisition and decentralized motion governed by local interaction rules. A normalized point cloud visibility metric was introduced to quantify mesh coverage without requiring explicit 3D reconstruction, enabling consistent evaluation between the two methods.

Key Findings: Results demonstrated that swarm-based methods achieved high surface coverage within shorter mission times and reduced travel distances, but required substantially higher image captures. In contrast, NBV planning attained similar reconstruction quality levels using significantly fewer observations, at the cost of longer execution times and increased trajectory lengths. Further analysis showed diminishing returns with increasing swarm size, NBV iterations, and number of drones. These findings highlight trade-offs between flight time, trajectory and capture efficiency, motivating future hybrid approaches that combine the strengths of both paradigms.

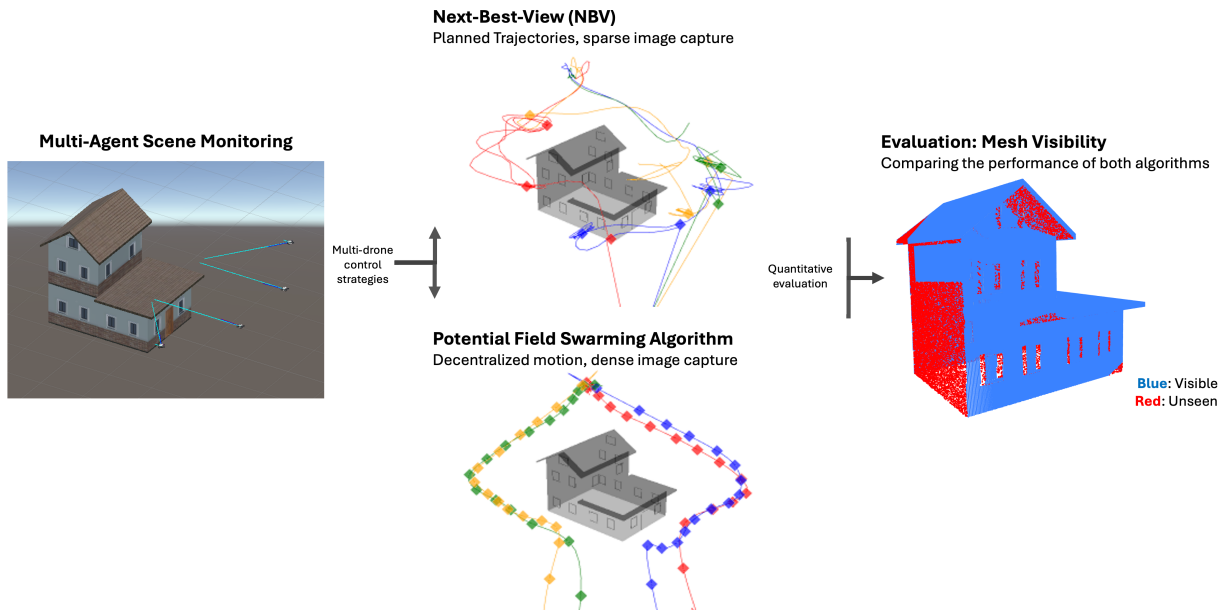


Fig. . High-level overview of the experimental framework. Multiple drones monitor a 3D structure using two multi-agent strategies: next-best-view (NBV) planning, which selects sparse, information-driven viewpoints through explicit trajectory planning, and a potential-field-based swarming algorithm, which produces decentralized motion and dense image capture. Both approaches are evaluated using a mesh visibility metric that quantifies surface coverage of a captured point cloud from accumulated observations, enabling direct comparison.

Next-Best-View Aerial Swarms for Multi-Viewpoint Monitoring

Advaith Sriram, Benjamin Jarvis, Dario Floreano, *Fellow, IEEE*

Abstract—Multi-drone systems are increasingly used for autonomous scene monitoring and 3D reconstruction, where efficient viewpoint selection and coverage remain key challenges. This work presents a comparative study of swarm-based exploration and Next-Best-View (NBV) planning for multi-drone systems in simulated environments. A normalized point cloud visibility metric is introduced to quantify mesh coverage without requiring explicit 3D reconstruction, enabling consistent evaluation between the two methods. Swarm-based exploration was implemented using periodic depth image acquisition and decentralized motion governed by local interaction rules, while NBV planning leveraged partial point cloud completion to iteratively select informative viewpoints. Experiments were conducted on diverse textured house models from the House3K dataset, with multiple drones operating in parallel. Results demonstrated that swarm-based methods achieved high surface coverage within shorter mission times and reduced travel distances, but required substantially more image captures. In contrast, NBV planning attained comparable reconstruction quality using significantly fewer observations, at the cost of longer execution times and increased trajectory lengths. Further analysis showed diminishing returns with increasing swarm size, NBV iterations, and number of drones. These findings highlight trade-offs between flight time, trajectory and capture efficiency, motivating future hybrid approaches that combine the strengths of both paradigms.

Index Terms—Next-best-view, aerial swarms, mesh visibility, point cloud completion.

I. INTRODUCTION

AUTONOMOUS aerial drones have become an increasingly important tool for monitoring, inspection, mapping, and exploration of complex environments [1] [2]. In particular, multi-robot swarms offer significant potential due to their scalability, robustness, and ability to provide simultaneous multi-viewpoint observations of a scene [3]. By distributing sensing and control across multiple agents, swarms can cover larger areas, reduce task completion time, and offer redundancy compared to single-agent systems. In parallel, Next-Best-View (NBV) planning [4] has emerged as an effective strategy for actively selecting camera viewpoints that maximise information gain about a scene based on partial knowledge. NBV approaches are widely used in applications such as 3D reconstruction [5], inspection [6], and exploration [7], where intelligently chosen viewpoints can substantially improve data quality and efficiency.

Traditionally, NBV planning has been studied in the context of single-agent systems, where a single drone or robotic sensor iteratively predicts where to move next in order to observe the most informative or previously unseen regions

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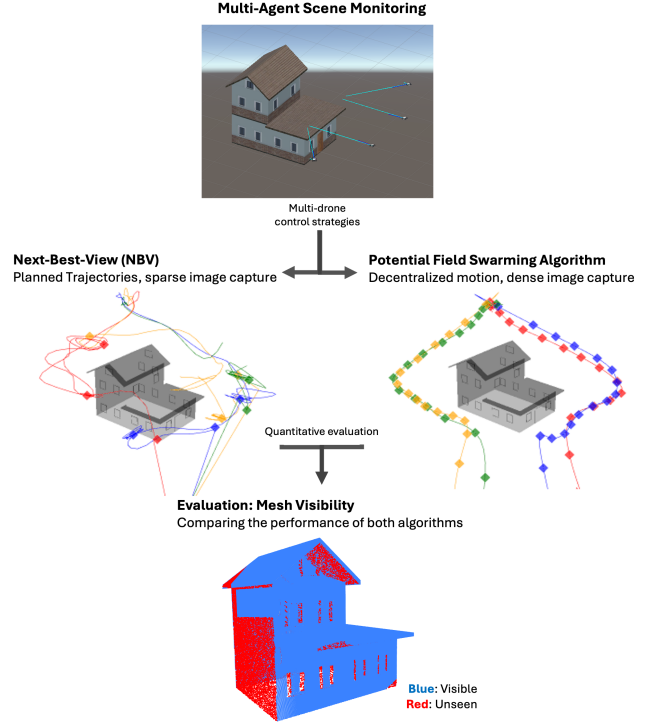


Fig. 1. High-level overview of the experimental framework. Multiple drones monitor a 3D structure using two multi-agent strategies: next-best-view (NBV) planning, which selects sparse, information-driven viewpoints through explicit trajectory planning, and a potential-field-based swarming algorithm, which produces decentralized motion and dense image capture. Both approaches are evaluated using a mesh visibility metric that quantifies surface coverage of a captured point cloud from accumulated observations, enabling direct comparison.

of a scene, or a large-scale environment itself [8]. The core idea is to evaluate candidate viewpoints and select the one that maximises information gain or reduces uncertainty [9], [10]. This paradigm has proven highly effective for tasks including mapping, inspection, and reconstruction [11], as different viewpoints reveal complementary information about an object or environment. More recently, research has begun to extend NBV planning to multi-agent settings, in which multiple drones cooperate by sharing observations or predictions to improve coverage efficiency and reduce redundancy.

Multi-agent NBV methods span a range of modelling and optimisation strategies depending on the sensing modality and reconstruction objective. Some approaches rely on neural uncertainty estimation [9] or surface-driven estimation [8]. Others, such as MAP-NBV [12], operate directly on partial geometric observations, using point cloud fusion and comple-

tion models to reason about unobserved regions of the scene. In this framework, each agent contributes local observations, enabling joint information gain maximisation across the team without explicit path planning or trajectory optimisation.

Separately, swarming algorithms provide a well-established framework for coordinating large numbers of drones using simple, decentralised control rules. Classical swarming approaches are commonly based on potential field formulations, in which each agent computes its motion from locally defined interaction terms such as attraction, repulsion, alignment, and obstacle avoidance [13]. These methods are computationally efficient, scalable, and robust, making them particularly well-suited for real-time multi-drone control. A key advantage of swarm-based systems is their inherent redundancy, as no single agent is critical to overall system performance, improving fault tolerance and resilience to individual agent failures [14].

Through inter-agent distance regulation and obstacle avoidance, swarming algorithms naturally drive agents to spread out while maintaining cohesion around a shared region of interest. This behaviour is conceptually similar to multi-agent NBV, as multiple agents collectively observe different parts of the environment. However, unlike NBV-based approaches, classical swarming algorithms do not explicitly optimise for information gain or scene visibility. As a result, despite their emergent multi-viewpoint behaviour, the effectiveness of classical swarming algorithms in observing differing viewpoints of a scene has not been systematically or quantitatively compared to NBV-based planning. This motivates the central goal of this project: to evaluate how well a potential-field-based swarming algorithm performs on next-best-view tasks, and to compare its scene visibility and operational costs against an explicit multi-agent NBV approach that directly optimises information gain.

II. METHOD

This work compares a multi-agent Next-Best-View (NBV) planner and a classical swarming algorithm for multi-viewpoint scene monitoring. Specifically, MAP-NBV is reproduced as a representative multi-agent NBV method, and the Olfati-Saber potential-field algorithm is implemented as a decentralised swarming baseline.

A. Multi-Agent Next-Best-View Planning (MAP-NBV)

MAP-NBV, outlined in Algorithm 1 is a decentralised, prediction-guided NBV framework designed for active 3D object reconstruction using multiple agents. Each drone incrementally observes the scene using onboard RGB-D sensors and shares its observations with neighbouring agents, enabling coordinated viewpoint selection that maximises information gain while reducing redundant coverage.

At each iteration, every drone captures a depth image of the scene. A segmentation mask is applied to isolate the object of interest, and the resulting segmented depth map is reprojected into a partial 3D point cloud. Point clouds from neighbouring agents are transformed into a common global reference frame and fused to form a shared partial observation of the scene. While the original MAP-NBV formulation performs object segmentation in the RGB domain, this

Algorithm 1 Prediction-Guided Multi-Agent NBV (MAP-NBV: Project Implementation)

Require: Number of agents N , initial drone poses $\{\mathbf{T}_i^0\}_{i=1}^N$

Require: Camera intrinsics $\mathcal{K}$, information gain threshold λ

Ensure: Next-best-view pose for each agent

```

1: Initialize global point cloud  $\mathcal{P} \leftarrow \emptyset$ 
2: for each planning iteration do
3:   for each agent  $i \in \{1, \dots, N\}$  in parallel do
4:     Capture depth image  $D_i$ 
5:     Reproject  $D_i$  into local point cloud  $\mathcal{P}_i$ 
6:     Transform  $\mathcal{P}_i$  to global frame using  $\mathbf{T}_i$ 
7:   end for
8:   Fuse all  $\mathcal{P}_i$  into shared point cloud  $\mathcal{P}$ 
9:   Predict completed point cloud  $\hat{\mathcal{P}} \leftarrow \text{PoinTr}(\mathcal{P})$ 
10:  Compute centroid  $\mathbf{c}$  of  $\hat{\mathcal{P}}$ 
11:  Generate candidate viewpoints  $\mathcal{V}$  around  $\mathbf{c}$ 
12:  for each agent  $i = 1$  to  $N$  (sequentially) do
13:     $\mathcal{V}_{\text{valid}} \leftarrow \emptyset$ 
14:    for each  $v \in \mathcal{V}$  do
15:      Compute information gain  $IG(v)$ 
16:      if  $IG(v) \geq \lambda$  then
17:        Add  $v$  to  $\mathcal{V}_{\text{valid}}$ 
18:      end if
19:    end for
20:    Select  $v_i^* \leftarrow \arg \min_{v \in \mathcal{V}_{\text{valid}}} \|\mathbf{p}_v - \mathbf{p}_i\|$ 
21:    Assign  $v_i^*$  to agent  $i$ 
22:  end for
23:  Move all agents to assigned viewpoints
24: end for
```

work leverages privileged segmentation information available in simulation to obtain depth-based object masks. This choice ensures consistent object isolation, allowing the evaluation to focus on viewpoint selection and multi-agent coordination rather than segmentation performance.

To approximate the unobserved regions of the object, the fused partial point cloud is passed to a point cloud completion network based on PoinTr [15], to predict a complete object point cloud. The predicted point cloud is not treated as the ground truth, but rather serves as a geometric prior that enables the planner to reason about unseen surfaces during viewpoint selection. The original MAP-NBV uses an improved PoinTr-C [16] model. However, this work uses the standard PoinTr model due to its simpler integration and comparable performance for the considered object classes.

Candidate viewpoints are generated based on the predicted object geometry. Specifically, the centroid of the predicted point cloud is computed, and candidate camera poses are sampled on a set of concentric horizontal rings around this centroid at multiple radii and elevations. Each ring consists of uniformly spaced viewpoints with a fixed angular resolution, resulting in a discrete set of candidate poses distributed around the object. In total, 60 candidate viewpoints are generated per planning iteration (five rings with 12 viewpoints each), further explained in Appendix A. This strategy ensures that candidate viewpoints are dynamically adapted to the current object estimate rather than relying on a predefined viewpoint

set. For each candidate viewpoint, the expected information gain is computed as the number of newly visible surface points relative to previously observed regions. Following the MAP-NBV strategy, candidate viewpoints whose information gain exceeds a predefined threshold ($\lambda = 0.75$) are considered valid. Among these valid candidates, the viewpoint closest to the current drone position is selected to minimise control effort while maintaining informative coverage. Agents select viewpoints using a sequential greedy algorithm while accounting for previously assigned viewpoints to reduce inter-agent redundancy.

B. Swarming-Based Multi-Agent Control (Olfati-Saber)

In contrast to NBV-based planning, swarming algorithms coordinate multiple agents using local interaction rules rather than explicit optimisation of information gain. In this work, the Olfati-Saber algorithm is employed as a representative potential-field-based swarming method for multi-agent control.

The Olfati-Saber algorithm computes a desired acceleration for each agent as the weighted summation of interaction forces that regulate inter-agent spacing, enforce velocity alignment, collision and obstacle avoidance. These forces are derived from artificial potential fields and depend only on local neighbour information within a finite sensing radius. As a result, the control law is fully distributed, scalable with the number of agents, and well-suited for real-time multi-drone coordination. The mathematical formulation of the Olfati-Saber control law is provided in Appendix B.

While classical swarm behaviours prioritize formation stability and collision avoidance, they do not explicitly constrain sensor orientation or viewpoint coverage. To adapt the swarm for multi-viewpoint monitoring, a convex hull is computed over the current drone positions to identify the boundary agents of the swarm at each control iteration. The centroid of this hull provides a geometric center of the swarm configuration, which approximates the object location when the swarm surrounds the target structure.

In prior internal laboratory implementations, boundary agents were oriented outward from the convex hull to maximise spatial dispersion and environmental coverage. In this work, this orientation strategy is inverted: boundary drones orient their yaw inward, towards the centroid of the convex hull. This modification allows for simultaneous observation of a common object while preserving the decentralised swarm dynamics (see Appendix C for further details).

C. System Architecture and Simulation Pipeline

All experiments are conducted in a Unity-based simulation environment, where multiple aerial drones equipped with depth cameras observe textured 3D object models. Each drone operates as an independent agent while exchanging data with an external processing module through a shared memory interface. This design decouples perception and decision-making from simulation, enabling flexible integration of different control and planning algorithms.

For NBV-based experiments, a shared memory space is used for storing the depth images and receiving viewpoint positional commands. These discrete viewpoint commands are generated

Algorithm 2 Swarm-Based Multi-Agent Control (Olfati-Saber)

Require: Number of agents N

Require: Camera intrinsics $\mathcal{K}$, sampling interval Δt

Ensure: Fused point cloud $\mathcal{P}$ collected during migration

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1: Initialize swarm agents at  $\mathbf{p}_{\text{start}}$ 
2: Initialize global point cloud  $\mathcal{P} \leftarrow \emptyset$ 
3: Initialize timer  $t \leftarrow 0$ 
4: while swarm has not reached  $\mathbf{p}_{\text{goal}}$  do
5:   for each agent  $i \in \{1, \dots, N\}$  do
6:     Update motion using Olfati-Saber interaction rules
7:   end for
8:   if  $t \bmod \Delta t = 0$  then
9:     for each agent  $i \in \{1, \dots, N\}$  in parallel do
10:      Capture depth image  $D_i$ 
11:      Reproject  $D_i$  into local point cloud  $\mathcal{P}_i$ 
12:      Transform  $\mathcal{P}_i$  to global frame using  $\mathbf{T}_i$ 
13:    end for
14:    Fuse all  $\mathcal{P}_i$  into global point cloud  $\mathcal{P}$ 
15:   end if
16:   Increment time  $t$ 
17: end while

```

periodically after each NBV iteration. Specifically, the planner processes the accumulated observations, selects a new target viewpoint for each drone, and sends updated position commands back through shared memory. The drones then move to these positions before the following NBV iteration begins. In contrast, for swarming-based experiments, no explicit control commands are sent back, and the depth images are stored locally. All drones are spawned at a common start location and are assigned a shared migration goal, Table I. Their motion emerges solely from local interaction rules, including inter-agent distance regulation and obstacle avoidance, while progressing toward the migration point. The depth camera observations are used exclusively for evaluation purposes and do not influence drone motion during swarm execution.

This unified pipeline enables direct comparison between NBV-based planning and swarming-based control under identical sensing, simulation, and evaluation conditions. A schematic overview of the system architecture and data flow is provided in Figure 2.

D. Visibility Metric

To quantitatively evaluate scene observability without requiring full 3D reconstruction, a mesh surface visibility metric based on point cloud coverage is employed. The metric measures how much of the ground-truth object surface is represented by the reconstructed point cloud accumulated from drone depth measurements. For each point on the ground-truth object surface, the distance to the nearest point in the reconstructed point cloud is computed. A surface point is considered visible if this distance lies below a predefined proximity threshold. The overall visibility score is defined as the fraction of ground-truth surface points that satisfy this visibility condition.

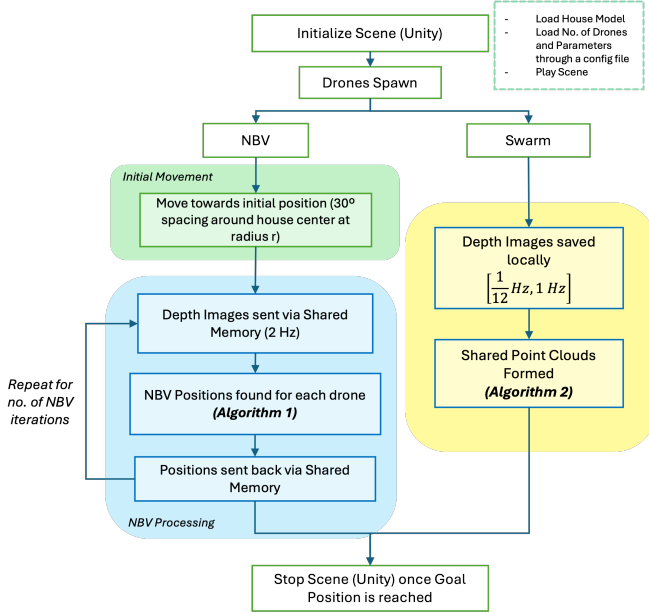


Fig. 2. High-level system architecture and simulation pipeline implemented in Unity for both MAP-NBV and swarm-based methods. The scene is initialised by loading a textured 3D house model and simulation parameters from a configuration file. In the NBV pipeline, positional commands are sent to the drones through shared memory, while in contrast, the swarm pipeline does not receive external motion commands.

The proximity threshold of 0.1 meters is defined in Unity’s world coordinates and reflects a tolerance on surface approximation accuracy. When point clouds are normalized for processing, the threshold is scaled accordingly, ensuring that the visibility metric remains consistent across objects of varying sizes. The mathematical definition of the visibility criterion and coverage computation is provided in Appendix D. The final visibility score is reported as a percentage of the total ground-truth surface area.

III. RESULTS

A. Experimental Setup

Experiments were conducted using 3D house models from the House3K dataset [5], which contains over 3,000 textured residential structures of varying shapes and architectural complexity. From this dataset, five house models, Figure 3, were randomly selected for evaluation, ensuring diversity in size, geometry, and architectural features. These models were specifically designed for NBV and 3D reconstruction tasks, making them well-suited for evaluating multi-viewpoint monitoring strategies. The list of experimental parameters is found in Table I.

B. Drone Trajectories and Data Collection

Figure 4 illustrates the three-dimensional trajectories of four drones operating under the MAP-NBV planning strategy and the swarming-based control strategy, respectively. In both cases, drones start from a common spawn location and terminate at a predefined goal position, while observing the same house model.

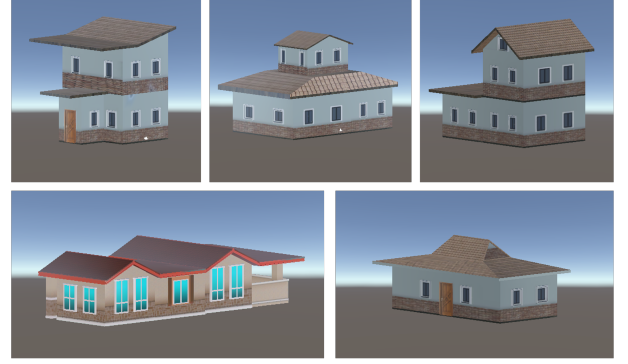


Fig. 3. Textured 3D house models used in the experimental evaluation, selected from the House3K [5] dataset. The dataset consists of diverse residential building models specifically designed for next-best-view and active perception research. For this study, a subset of five house models was randomly chosen while ensuring variability in geometric and visual properties, including overall size, building height, roof structure, and surface textures, providing a robust assessment of scene visibility performance under varying structural conditions.

TABLE I
EXPERIMENTAL PARAMETERS FOR NBV-BASED AND SWARMING-BASED EVALUATIONS

Category	Parameter	NBV	Swarm
Environment	Dataset	House3K	House3K
Agents	Number of drones (N)	[1, 9]	[1, 9]
Initialization	Spawn (x, z)	(−50, −50)	(−50, −50)
	Target (x, z)	(50, 50)	(50, 50)
View Planning	NBV iterations	[0, 5]	–
	Initial viewpoints	30° spacing	–
Sensing	Camera type	Depth	Depth
	Sampling interval (s)	–	[1, 12]
Evaluation	Houses tested	5 models	5 models

Under the MAP-NBV strategy, drones follow longer and more irregular trajectories around the object. Each agent repeatedly relocates to discrete NBV positions selected by the NBV planner, resulting in significant lateral and vertical motion around the structure. Across all agents, the total distance travelled and overall mission duration are noticeably higher than in the swarming case. The NBV configuration performs a fixed number of planning iterations (two), resulting in a total of 12 depth images collected (three captures per drone). In contrast, the swarming-based approach produces smoother and more direct trajectories from the start position to the goal position. Drones maintain formation while navigating around the object using local interaction rules, resulting in shorter path lengths and reduced traversal time. However, depth images are captured periodically throughout the motion, leading to a higher total number of observations. For the experiment shown, each of the four drones captures one depth image every five seconds, yielding a total of 72 images. These trajectories illustrate qualitative differences in motion and execution time between the two strategies. The number of image captures shown corresponds to the selected sampling configuration and is used here for illustrative consistency rather than to imply a required capture frequency.

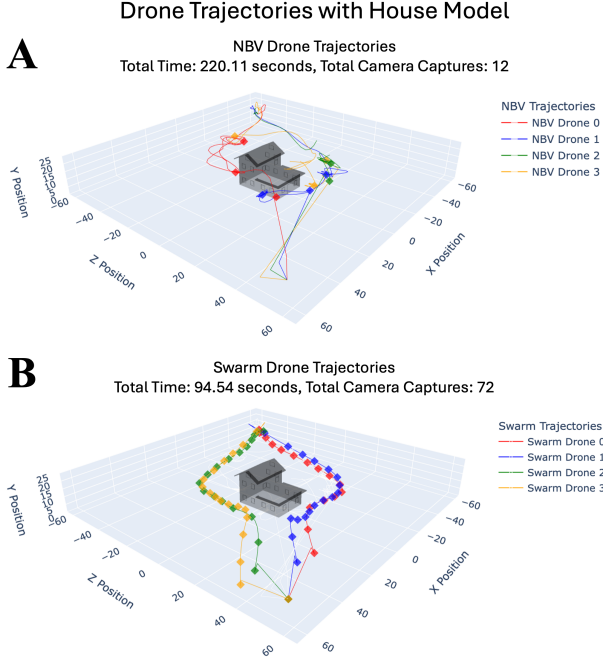


Fig. 4. 3-D drone trajectories for MAP-NBV and swarming-based control around a house model, with four drones. The diamond marker represents where a capture was taken. **A)** MAP-NBV trajectories with two planning iterations following the initial observation, resulting in three depth captures per drone (12 images total). **B)** Swarming-based trajectories capturing depth images periodically every 5 seconds during continuous motion. The swarm traversal duration was 94.54 seconds, resulting in a total of 72 depth images. While MAP-NBV produces longer and more complex trajectories with higher traversal time (220.11 seconds), it requires significantly fewer image captures compared to the swarming-based approach.

C. Qualitative Coverage Analysis

Figure 5 provides a qualitative comparison of surface observability for swarm-based exploration and NBV-based planning under the same experimental configuration as Figure 4. The visualization depicts ground-truth surface points that were either observed (in blue) or unseen (in red). The swarm-based approach achieves high overall coverage (96.60%), with only 3.40% of surface points remaining unobserved. The unseen regions are primarily confined to geometrically challenging structures, such as roof overhangs and recessed surfaces, which are difficult to observe from the predominantly horizontal viewing angles produced by the swarm trajectories. These missed regions are spatially localized and correspond to surfaces that remain occluded throughout the swarm traversal. In contrast, the NBV-based approach results in substantially lower total coverage (77.42%), with 22.58% of the surface remaining unseen. Most notably, an entire face of the house remains unobserved in the NBV case.

D. Effect of Swarm Size and Camera Sampling Frequency

The mesh visibility achieved by the swarming-based approach as a function of the number of drones is shown in Figure 6. Mesh visibility increases sharply when moving from a single drone to a two-drone swarm, rising from approximately 40% to over 80% coverage across all camera sampling intervals. Adding a third drone further improves visibility to

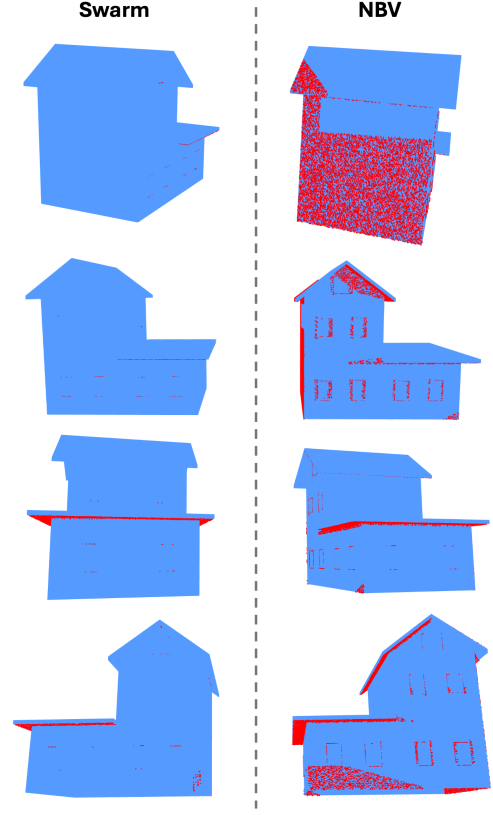


Fig. 5. Binary observability visualization of unseen surface regions for **swarm-based exploration (left)** and **NBV-based planning (right)** using four drones. Ground-truth surface points are shown in blue, while unseen points are shown in red. The swarm configuration (4 drones, 5 s sampling interval) achieves higher overall coverage (96.60%) but leaves certain architectural features, such as roof overhangs, unseen. The NBV configuration (4 drones, 2 iterations) results in lower total coverage (77.42%).

TABLE II
MEAN MESH VISIBILITY FOR THE SWARMING-BASED APPROACH ACROSS DIFFERENT CAMERA SAMPLING INTERVALS USING A FIXED SWARM SIZE OF FOUR DRONES. RESULTS ARE AVERAGED OVER ALL HOUSE MODELS.

Sampling Interval (s)	Mean Visibility (%) $\pm$ Std
1	88.63 $\pm$ 6.89
2	88.00 $\pm$ 6.93
3	88.02 $\pm$ 6.43
4	87.92 $\pm$ 7.78
5	87.78 $\pm$ 6.37
6	86.67 $\pm$ 7.10
7	87.48 $\pm$ 6.52
8	86.61 $\pm$ 7.75
9	86.97 $\pm$ 7.60
10	86.40 $\pm$ 8.67
11	85.57 $\pm$ 8.48
12	85.98 $\pm$ 8.15

approximately 85%, after which gains become progressively smaller. Across all sampling intervals, the mean visibility saturates at approximately 85-90% for swarm sizes of five drones and above. The mean visibility, shown by the black markers, indicate that increasing the number of drones beyond this point yields minimal improvement in overall surface coverage. The lower plot illustrates the marginal improvement in visibility achieved by adding each additional drone.

In addition to varying swarm size, the effect of camera sampling interval on mesh visibility was evaluated for the

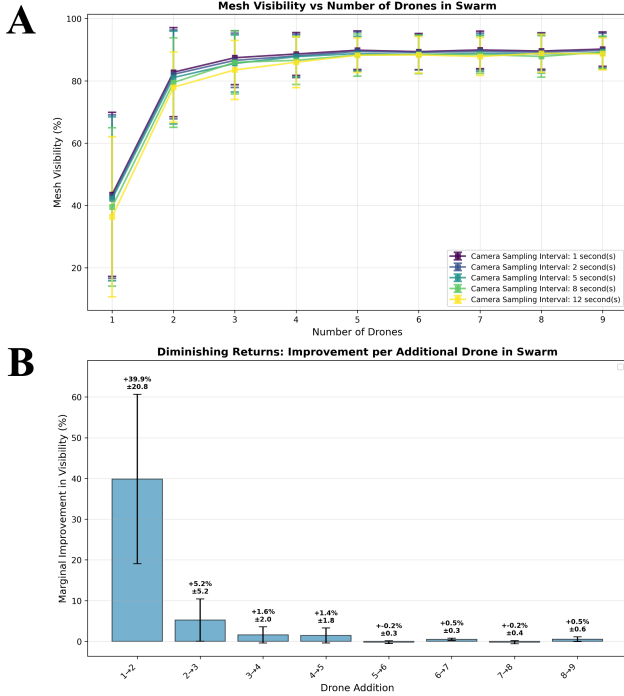


Fig. 6. Mesh visibility as a function of the number of drones in the swarming-based approach. **A)** Mean mesh visibility across different camera sampling intervals (1, 2, 5, 8, and 12 seconds), with standard deviation across houses. **B)** Marginal improvement in mesh visibility obtained by adding an additional drone to the swarm, computed as the difference in mean visibility between consecutive swarm sizes. The results illustrate a rapid increase in visibility when increasing the swarm size from one to two drones, followed by diminishing returns as additional drones are added.

swarming-based approach using a fixed swarm size of four drones. Sampling intervals ranging from 1 second to 12 seconds were tested. Table II reports the mean mesh visibility and standard deviation averaged over all house models for each sampling interval. The results show no significant reduction in visibility as the sampling interval increases. Mean visibility remains within a narrow range of approximately 85-89% across all tested intervals, with overlapping standard deviations. This indicates that, for the swarming-based approach, overall scene coverage is largely insensitive to camera sampling frequency within the tested range.

E. Effect of Next-Best-View (NBV) Iterations and Drone Count

Figure 7 illustrates the marginal improvement in mesh visibility obtained from successive NBV iterations. Results are averaged across all drone counts (1 through 9) and house models. A large increase in visibility is observed between the initial observation and the first NBV iteration, with an average improvement of approximately 13.7%. Subsequent iterations yield progressively smaller gains, with the marginal improvement decreasing to below 2% after three iterations. This trend indicates diminishing returns in surface visibility as the number of NBV iterations increases.

Figure 8 shows mesh visibility as a function of the number of drones for different NBV iteration counts, along with the marginal improvement achieved by adding additional drones. Visibility increases consistently as the number of drones

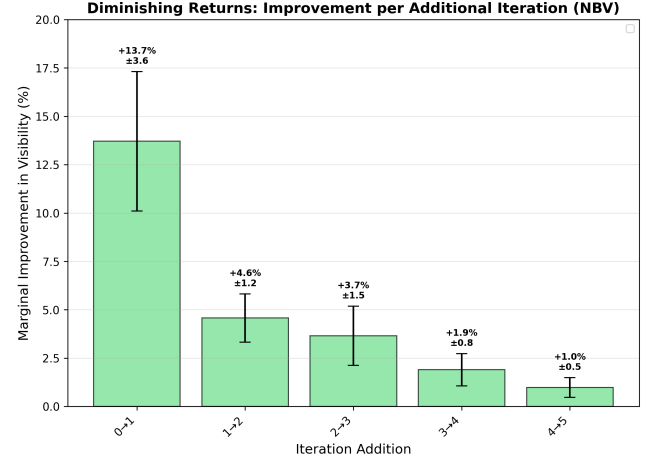


Fig. 7. Marginal improvement in mesh visibility obtained from successive Next-Best-View (NBV) planning iterations. The plot shows the average increase in surface visibility when progressing from one iteration to the next, with error bars indicating standard deviation across drone counts (1 to 9) and houses. A large gain is observed after the first NBV iteration, followed by diminishing returns in later iterations, indicating that most informative viewpoints are identified early in the planning process.

grows, particularly for lower drone counts. However, the marginal gain per additional drone decreases beyond approximately five to six drones, with improvements falling below 2-3% for higher swarm sizes.

IV. DISCUSSION

A. Trajectory Efficiency and Image Capture Trade-offs

The trajectory differences observed between MAP-NBV and the swarming-based approach stem directly from their underlying control methods. MAP-NBV explicitly optimises viewpoint selection to maximise information gain, which frequently requires drones to reposition around the object to access previously unseen surfaces. As a result, agents execute longer, non-smooth trajectories that prioritise coverage over travel efficiency. Although this leads to increased flight time and distance travelled, it enables the system to acquire informative observations using a relatively small number of strategically selected images. In contrast, in the swarming-based approach, drones follow local interaction rules while progressing toward a global migration goal. This results in more direct and coordinated motion, reducing both traversal time and total distance travelled. However, because sensing is decoupled from planning, the swarm compensates by collecting depth observations at a higher temporal frequency. Consequently, the swarm accumulates a larger number of images despite executing shorter trajectories.

This contrast highlights a fundamental trade-off between motion efficiency and sensing strategy. MAP-NBV favours sparse but information-rich observations at the cost of increased maneuvering, whereas swarming favours continuous observation during efficient motion but lacks explicit control over viewpoint informativeness. These differences motivate a quantitative evaluation of scene visibility, presented in subsequent subsections, to determine how effectively each approach

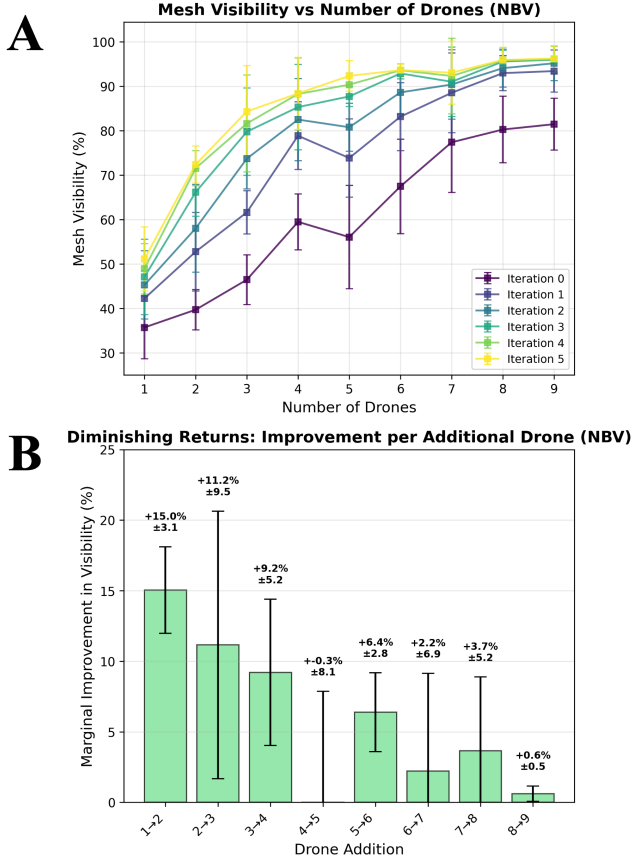


Fig. 8. Mesh visibility achieved by the NBV algorithm as a function of the number of drones. **A)** Mesh visibility for different NBV iteration counts with standard deviation across houses. **B)** Marginal improvement in visibility obtained by adding each additional drone. While increasing the number of drones improves overall visibility, the marginal benefit decreases beyond approximately five to six agents, indicating diminishing returns due to overlapping viewpoints and increased redundancy.

observes the object surface under their respective sensing and motion constraints.

B. Coverage Quality

Although swarm-based exploration achieves higher overall mesh surface visibility, the qualitative results reveal important differences in how coverage is distributed across the scene. As shown in Figure 5, the swarm-based approach leaves only a small fraction of the surface unobserved (3.40%). These missed regions are spatially localized and primarily correspond to geometrically challenging architectural features such as roof overhangs. While limited in extent, these regions remain completely unobserved due to persistent occlusions and the predominantly horizontal viewing angles induced by the swarm trajectories.

In contrast, the NBV-based approach shows lower overall visibility (77.42%), with 22.58% of the surface remaining unseen. Most notably, an entire face of the house is missed under the NBV configuration. This failure reflects the limitations in the predictive component of the NBV pipeline. In particular, the PoinTr shape completion model used to guide the viewpoint selection is not trained on architectural

structures, and does not accurately predict the full geometry of the house from partial observations. As a result, regions that are poorly represented in the predicted shape are not prioritized by the NBV planner, leading to the coverage gaps.

These results indicate that under the current configuration, swarm-based exploration provides more uniform surface observability, while the effectiveness of NBV-based planning is dependent on the quality of the underlying shape prediction. Finetuning the predictive model or incorporating architectural priors would help improve the performance of the NBV planner.

C. Diminishing Returns in Swarm-Based Coverage

The results demonstrate diminishing returns with respect to swarm size in the swarming-based approach. The substantial improvement observed when increasing the swarm size from one to two drones can be attributed to the introduction of multi-viewpoint observations, which significantly reduce occlusions and blind spots inherent to single-agent sensing.

Beyond two or three drones, additional agents increasingly observe regions that are already partially covered, leading to overlapping fields of view and redundant observations. As a result, the marginal contribution of each additional drone decreases rapidly, and mesh visibility saturates at approximately 90%. This saturation effect persists across all tested camera sampling intervals, indicating that swarm size, rather than sensing frequency, is the dominant factor influencing coverage beyond a certain point. These findings suggest that while swarming-based motion naturally enables distributed scene observation, it does not explicitly optimise for coverage efficiency. Without a mechanism to actively select informative viewpoints, additional drones primarily increase redundancy rather than uncovering new surface regions. This behaviour contrasts with the design philosophy of NBV-based approaches, which explicitly reason about information gain to reduce redundant observations and prioritise informative viewpoints. From a practical perspective, these findings reveal an important trade-off: increasing the number of drones improves robustness and redundancy but yields diminishing improvements in visibility while increasing system complexity, sensing load, and data volume.

The weak dependence of mesh visibility on camera sampling interval suggests that spatial coverage, rather than sensing frequency, is the primary factor governing observability in the swarming-based approach. As drones traverse the environment and distribute themselves around the object, the majority of the visible surface is captured through spatial diversity rather than dense temporal sampling. Even at low sampling frequencies, the continuous motion of the swarm allows drones to observe new surface regions over time. Additional captures taken at higher frequencies tend to be redundant, as successive viewpoints are spatially similar. This behaviour explains why reducing the sampling interval does not lead to a significant decrease in mesh visibility. These findings further emphasise the role of swarm size and spatial configuration in determining coverage performance. While higher sampling frequencies increase data volume and processing load, they provide limited benefit in terms of additional surface visibility.

D. Next-Best-View (NBV) Performance Analysis

The NBV results exhibit diminishing returns along two complementary axes: the number of planning iterations and the number of agents. The substantial improvement observed during the first NBV iteration reflects the algorithm’s ability to reposition drones toward previously unseen regions based on the predicted object geometry. Once the most informative viewpoints have been exploited, additional iterations primarily refine already observed surfaces, resulting in progressively smaller gains in visibility. A similar effect is observed when increasing the number of drones. While adding agents initially improves coverage by enabling simultaneous observation from multiple viewpoints, the benefit of additional drones decreases as spatial redundancy increases. Beyond a moderate number of agents, newly added drones tend to observe regions already covered by existing viewpoints, especially given the finite surface area of the object.

Together, these results highlight a fundamental trade-off in NBV-based systems between coverage, coordination complexity, and efficiency. While NBV planning enables efficient information acquisition with relatively few observations, both excessive iteration counts and large agent counts provide limited additional benefit.

V. FUTURE WORK

While this work provides a quantitative comparison between swarm-based and Next-Best-View (NBV) strategies for multi-drone scene observability, several avenues remain to improve the robustness and generality of the framework. The current NBV pipeline relies exclusively on depth data and ground-truth object masks to generate partial point clouds. In contrast, the original MAP-NBV formulation [12] integrates RGB-based semantic segmentation with depth sensing to construct segmented depth maps. An attempt was made to incorporate RGB segmentation using the Segment Anything model [17], but the resulting segmentations were not sufficiently reliable for consistent reconstruction. Future work could explore more robust semantic or instance segmentation methods, including task-specific fine-tuning or multimodal fusion, to better align the implementation with the full MAP-NBV framework.

The PoinTr-based point cloud completion model [15] employed in this work was originally trained on everyday objects and is not specifically optimized for large-scale structures such as buildings. This mismatch likely limits the accuracy of full-shape prediction from partial observations. Fine-tuning the ShapeNet55-pretrained PoinTr model using datasets, such as the House3K dataset [5], could significantly improve reconstruction quality and NBV planning performance.

For the swarm-based approach, future work could further investigate the impact of camera sampling rate. While this study evaluated sampling intervals of up to one image every 12 seconds, extending this analysis to lower imaging rates (e.g., 20 or 30 seconds) would help identify the point at which reduced temporal sampling begins to significantly degrade surface observability. Such analysis would provide clearer guidance on the minimum sensing requirements for effective swarm-based exploration while reducing sensing and processing overhead.

An important direction for future investigation is the role of structural complexity in multi-agent monitoring. The house models considered in this work contain relatively simple geometry with limited concave regions. More complex architectural structures containing deep recesses, overhangs, and self-occlusions may exacerbate the limitations of swarm-based exploration, as potential-field-driven motion does not explicitly reason about occluded or hard-to-observe surfaces. In contrast, NBV-based methods are explicitly designed to target unobserved regions and may better adapt to such geometries. Systematically varying architectural complexity and quantifying its impact on coverage would provide deeper insight into the strengths and failure modes of each approach.

While NBV achieves high reconstruction quality with relatively few observations, it incurs longer mission times and increased travel distances. Swarm strategies, on the other hand, provide rapid coverage but require significantly more observations. These limitations suggest the need for a unified approach, combining NBV planning with swarm-based exploration. By employing NBV positions as waypoints for a decentralized swarm, the system could combine the strengths of both methods resulting in both informative viewpoints and efficient traversal across the scene.

VI. CONCLUSION

This work presented a comparative study of swarm-based exploration and Next-Best-View (NBV) planning for multi-drone scene observability and surface reconstruction. Experimental results demonstrate that swarm-based methods achieve high surface coverage within significantly shorter mission times and with reduced travel distances, albeit at the cost of substantially higher numbers of image captures. In contrast, NBV planning achieves comparable mesh visibility using far fewer observations, highlighting its efficiency in information acquisition, but at the expense of longer execution times and increased trajectory lengths.

Analysis of diminishing returns revealed that both approaches exhibit saturation effects. For swarm exploration, increasing the number of drones yields rapidly decreasing marginal gains beyond a moderate swarm size, while sampling interval variations have minimal impact on reconstruction quality within the tested range. Similarly, NBV performance shows diminishing improvements with additional planning iterations and increasing numbers of drones, indicating that most informative viewpoints are identified early and that excessive agent redundancy provides limited benefit. Together, these findings highlight an inherent trade-off between temporal efficiency, sensing redundancy, and planning complexity in multi-drone monitoring systems. By explicitly quantifying these trade-offs, this work provides insight into the strengths and limitations of both swarm-based and NBV-driven strategies and motivates future hybrid approaches that combine structured viewpoint selection with parallel exploration. Such methods hold promise for scalable, efficient, and high-fidelity monitoring of complex real-world environments.

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APPENDIX

A. Candidate Viewpoint Sampling Details

Candidate viewpoints are sampled on five concentric rings centred at the centroid of the predicted point cloud. Each ring contains 12 viewpoints, uniformly spaced at 30° intervals in azimuth, resulting in 60 discrete candidate camera poses per planning iteration. The ring radii are scaled relative to the

maximum extent of the predicted object (d_{max}), and vertical offsets are applied to sample viewpoints at different elevations above and below the object centroid.

TABLE III
CANDIDATE VIEWPOINT SAMPLING CONFIGURATION USED FOR NBV PLANNING.

Ring index	Radius scaling ($\times d_{max}$)	Vertical offset
1	0.8	+12.5
2	1.0	+10.0
3	1.2	+5.0
4	1.5	0.0
5	1.2	-5.0

B. Olfati-Saber Swarming Model

For each agent i , the Olfati-Saber model computes a desired acceleration $\mathbf{u}_i$ given by

$$\mathbf{u}_i = \sum_{j \in \mathcal{N}_i} \mathbf{f}_{\text{dist}}(\mathbf{p}_i, \mathbf{p}_j) + \sum_{j \in \mathcal{N}_i} \mathbf{f}_{\text{vel}}(\mathbf{v}_i, \mathbf{v}_j) + \mathbf{f}_{\text{obs}}(\mathbf{p}_i), \quad (1)$$

where $\mathbf{p}_i$ and $\mathbf{v}_i$ denote the position and velocity of agent i , and $\mathcal{N}_i$ is the set of neighbouring agents within a finite interaction radius.

The term $\mathbf{f}_{\text{dist}}(\cdot)$ enforces inter-agent distance regulation through an artificial potential function, $\mathbf{f}_{\text{vel}}(\cdot)$ promotes velocity consensus among neighbouring agents, and $\mathbf{f}_{\text{obs}}(\cdot)$ accounts for obstacle avoidance. The exact functional forms and parameters follow the original formulation by Olfati-Saber [13].

C. Boundary Drone Yaw Orientation Strategy

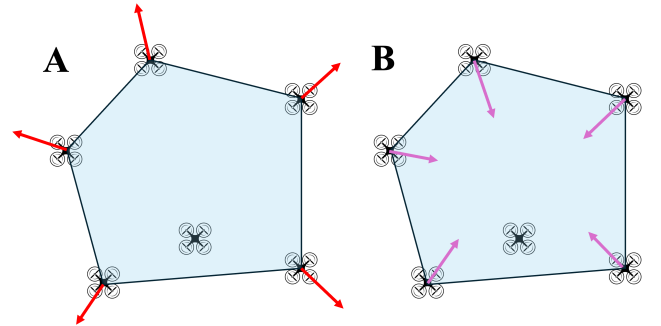


Fig. 9. Comparison of boundary drone yaw orientation strategies. **A)** Previous internal laboratory implementations, where drones located on the convex hull orient their yaw outward (red arrows) to maximise spatial dispersion and environmental coverage. **B)** Orientation strategy adopted in this work, where the yaw direction is inverted such that boundary drones orient inward toward the convex hull centroid (purple arrows). When encircling the target object in the scene, this configuration enables simultaneous observation of the target object while maintaining decentralised swarm behaviour.

D. Visibility and Coverage Metric Definition

Let $\mathcal{P}_{gt} = \{\mathbf{x}_j\}$ denote the ground-truth point cloud representing the object surface, and let $\mathcal{P}_{rec} = \{\mathbf{y}_i\}$ denote the reconstructed point cloud obtained from accumulated depth observations.

For each ground-truth point $\mathbf{x}_j \in \mathcal{P}_{gt}$, the Euclidean distance to its nearest neighbour in $\mathcal{P}_{rec}$ is computed:

$$d_j = \min_{\mathbf{y}_i \in \mathcal{P}_{rec}} \|\mathbf{x}_j - \mathbf{y}_i\|.$$

A ground-truth point is considered visible if $d_j < \tau$, where τ denotes the distance threshold. The coverage (visibility) metric is defined as:

$$\text{Coverage} = \frac{1}{|\mathcal{P}_{gt}|} \sum_{\mathbf{x}_j \in \mathcal{P}_{gt}} \mathbb{I}(d_j < \tau),$$

where $\mathbb{I}(\cdot)$ is the indicator function.

The threshold τ is defined in normalized object coordinates. A base threshold of $\tau = 0.1$ is used and scaled by the normalization factor applied to the reconstructed point cloud, ensuring scale-invariant evaluation across objects of varying dimensions.